

Problem Set 4

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1 Problem 1

Eggers and Hainmueller (2009) examine data on the wealth of UK MP candidates to estimate the effect of office-holding on wealth. This is an observational design, rather than an experiment. The raw data are available at <http://j.mp/32PHfFd>.

- a) Read the data into R and name it `mps`
- b) Create a new variable called `winner`. Set it equal to 1 if `margin` is greater than 0; set it to 0 if `margin` is less than zero.
- c) Consider `winner` to be a randomly assigned treatment. Estimate the causal effect of `winner` on `ln.net`, net worth at death of these candidates.
- d) Investigate whether the effect of `winner` varies by `party`.
- e) (Optional bonus question:) Replicate the “regression discontinuity design” of Eggers and Hainmueller (2009). That is, estimate four regression models of `ln.net` on `margin` (one for each combination of `party` and `winner`); predict the value of `ln.net` at `margin`= 0 from each model; calculate the difference in these predictions for each party and compare.

2 Problem 2

This exercise will give you practice creating and estimating treatment effects in blocked designs.

- a) Load the `blockTools` and bring the `x100` data into the workspace.
- b) Create a new variable `ra_tr` in the `x100` data, and randomly allocate half the units to treatment (1), half to control (0). (Set the random seed to make your allocation replicable.)
- c) Calculate the mean of `b1` for the treatment and control groups, and compare.
- d) Now block on `b1`, and assign treatment within blocks.
- e) Calculate the mean of `b1` for the treatment and control groups, and compare to part c).

References

Eggers, Andrew C., and Jens Hainmueller. 2009. “MPs for Sale? Returns to Office in Postwar British Politics.” *American Political Science Review* 103 (4): 513–33.